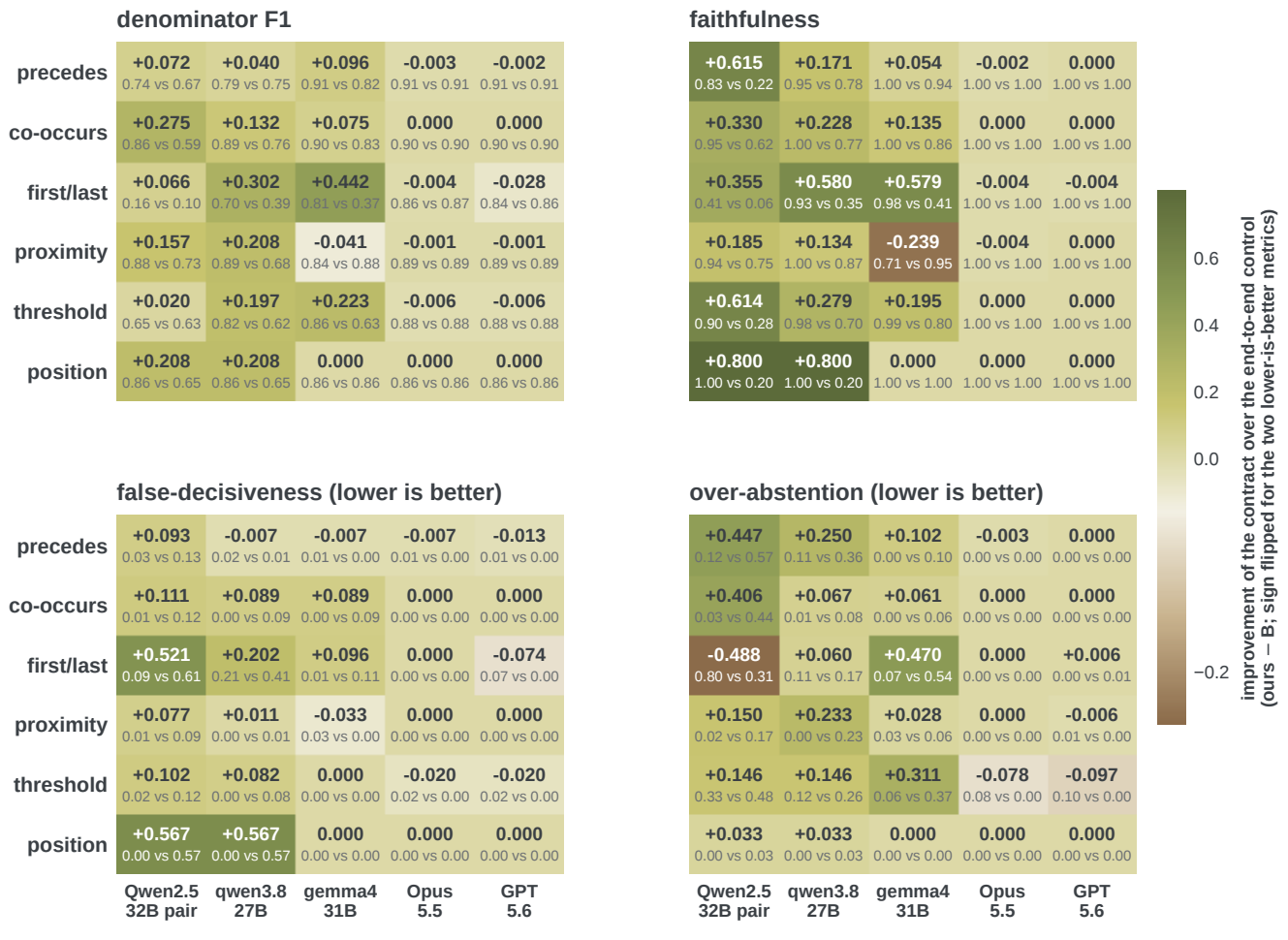
denominator F1
precedes
+0.072
0.74 vs 0.67
+0.040
0.79 vs 0.75
+0.096
0.91 vs 0.82
-0.003
0.91 vs 0.91
-0.002
0.91 vs 0.91
co-occurs
+0.275
0.86 vs 0.59
+0.132
0.89 vs 0.76
+0.075
0.90 vs 0.83
0.000
0.90 vs 0.90
0.000
0.90 vs 0.90
first/last
+0.066
0.16 vs 0.10
+0.302
0.70 vs 0.39
+0.442
0.81 vs 0.37
-0.004
0.86 vs 0.87
-0.028
0.84 vs 0.86
proximity
+0.157
0.88 vs 0.73
+0.208
0.89 vs 0.68
-0.041
0.84 vs 0.88
-0.001
0.89 vs 0.89
-0.001
0.89 vs 0.89
threshold
+0.020
0.65 vs 0.63
+0.197
0.82 vs 0.62
+0.223
0.86 vs 0.63
-0.006
0.88 vs 0.88
-0.006
0.88 vs 0.88
position
+0.208
0.86 vs 0.65
+0.208
0.86 vs 0.65
0.000
0.86 vs 0.86
0.000
0.86 vs 0.86
0.000
0.86 vs 0.86
faithfulness
+0.615
0.83 vs 0.22
+0.171
0.95 vs 0.78
+0.054
1.00 vs 0.94
-0.002
1.00 vs 1.00
0.000
1.00 vs 1.00
+0.330
0.95 vs 0.62
+0.228
1.00 vs 0.77
+0.135
1.00 vs 0.86
0.000
1.00 vs 1.00
0.000
1.00 vs 1.00
+0.355
0.41 vs 0.06
+0.580
0.93 vs 0.35
+0.579
0.98 vs 0.41
-0.004
1.00 vs 1.00
-0.004
1.00 vs 1.00
+0.185
0.94 vs 0.75
+0.134
1.00 vs 0.87
-0.239
0.71 vs 0.95
-0.004
1.00 vs 1.00
0.000
1.00 vs 1.00
+0.614
0.90 vs 0.28
+0.279
0.98 vs 0.70
+0.195
0.99 vs 0.80
0.000
1.00 vs 1.00
0.000
1.00 vs 1.00
+0.800
1.00 vs 0.20
+0.800
1.00 vs 0.20
0.000
1.00 vs 1.00
0.000
1.00 vs 1.00
0.000
1.00 vs 1.00
false-decisiveness (lower is better)
precedes
+0.093
0.03 vs 0.13
-0.007
0.02 vs 0.01
-0.007
0.01 vs 0.00
-0.007
0.01 vs 0.00
-0.013
0.01 vs 0.00
co-occurs
+0.111
0.01 vs 0.12
+0.089
0.00 vs 0.09
+0.089
0.00 vs 0.09
0.000
0.00 vs 0.00
0.000
0.00 vs 0.00
first/last
+0.521
0.09 vs 0.61
+0.202
0.21 vs 0.41
+0.096
0.01 vs 0.11
0.000
0.00 vs 0.00
-0.074
0.07 vs 0.00
proximity
+0.077
0.01 vs 0.09
+0.011
0.00 vs 0.01
-0.033
0.03 vs 0.00
0.000
0.00 vs 0.00
0.000
0.00 vs 0.00
threshold
+0.102
0.02 vs 0.12
+0.082
0.00 vs 0.08
0.000
0.00 vs 0.00
-0.020
0.02 vs 0.00
-0.020
0.02 vs 0.00
position
+0.567
0.00 vs 0.57
+0.567
0.00 vs 0.57
0.000
0.00 vs 0.00
0.000
0.00 vs 0.00
0.000
0.00 vs 0.00
Qwen2.5 32B pair
qwen3.8 27B
gemma4 31B
Opus 5.5
GPT 5.6
over-abstention (lower is better)
+0.447
0.12 vs 0.57
+0.250
0.11 vs 0.36
+0.102
0.00 vs 0.10
-0.003
0.00 vs 0.00
0.000
0.00 vs 0.00
+0.406
0.03 vs 0.44
+0.067
0.01 vs 0.08
+0.061
0.00 vs 0.06
0.000
0.00 vs 0.00
0.000
0.00 vs 0.00
-0.488
0.80 vs 0.31
+0.060
0.11 vs 0.17
+0.470
0.07 vs 0.54
0.000
0.00 vs 0.00
+0.006
0.00 vs 0.01
+0.150
0.02 vs 0.17
+0.233
0.00 vs 0.23
+0.028
0.03 vs 0.06
0.000
0.00 vs 0.00
-0.006
0.01 vs 0.00
+0.146
0.33 vs 0.48
+0.146
0.12 vs 0.26
+0.311
0.06 vs 0.37
-0.078
0.08 vs 0.00
-0.097
0.10 vs 0.00
+0.033
0.00 vs 0.03
+0.033
0.00 vs 0.03
0.000
0.00 vs 0.00
0.000
0.00 vs 0.00
0.000
0.00 vs 0.00
Qwen2.5 32B pair
qwen3.8 27B
gemma4 31B
Opus 5.5
GPT 5.6
0.6
0.4
0.2
0.0
−0.2
improvement of the contract over the end-to-end control
(ours − B; sign flipped for the two lower-is-better metrics)